

KIGALI INTL, Rwanda

WMO: 643870

Lat: 1.969S

Lon: 30.139E

Elev: 1481

StdP: 84.75

Time Zone: 2.00 (E02)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	14.9	15.2	6.8	7.4	24.8	8.0	8.0	23.9	8.5	21.7	6.8	22.5	1.2	320	0.266

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.8	30.2	18.6	29.4	18.6	28.9	18.5	21.4	26.0	21.1	25.7	20.7	25.4	3.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	17.8	22.4	20.0	17.6	22.3	19.2	16.8	21.8	69.9	25.9	68.5	25.9	67.3	25.6	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 6.6	5.5	4.9	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	10.4	22.4	0.9	0.7	9.7	22.9	9.2	23.3	8.7	23.7	8.0	24.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.8	22.3	22.3	21.8	21.3	21.4	21.4	21.7	22.5	22.3	22.0	21.2	21.5
	DBStd	1.27	1.31	1.47	1.44	1.18	0.91	0.87	1.09	1.16	1.20	1.32	1.10	1.16
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	4306	381	344	365	340	353	343	362	386	368	372	337	356
	CDD18.3	1266	122	111	107	90	95	93	103	128	118	113	87	98
	CDH23.3	6607	731	682	559	314	326	457	659	879	687	576	313	424
	CDH26.7	1102	172	177	106	21	15	27	81	176	135	106	30	58
Wind	WSAvg	2.0	1.9	1.9	1.8	1.9	1.8	1.9	2.0	2.3	2.3	2.1	1.9	1.8
Precipitation	PrecAvg	1193	86	85	148	154	119	35	28	60	95	125	144	113
	PrecMax	1592	177	240	253	241	205	76	94	104	204	265	225	238
	PrecMin	990	22	20	71	71	59	2	3	16	37	59	73	36
	PrecStd	137	34	41	38	36	35	20	18	22	33	37	38	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.0	31.9	30.2	28.2	28.0	28.2	29.3	30.8	30.2	30.1	28.8	29.2
		MCWB	18.2	18.4	19.8	20.4	19.9	18.6	17.5	17.5	18.3	18.5	20.2	19.4
	2%	DB	29.9	30.1	29.0	27.1	27.0	27.2	28.2	29.2	29.1	29.0	27.4	28.1
		MCWB	18.9	18.9	19.8	20.1	19.6	18.2	17.0	17.1	18.2	18.7	20.0	19.5
	5%	DB	28.9	28.9	27.9	26.2	26.1	26.9	27.6	28.3	28.2	27.9	26.6	27.1
		MCWB	18.9	18.9	19.8	20.0	19.5	18.0	16.9	17.0	18.0	18.7	19.9	19.5
	10%	DB	27.2	27.2	26.8	25.2	25.1	26.0	26.9	27.8	27.1	26.8	25.1	26.0
		MCWB	19.2	19.3	19.8	19.7	19.2	17.8	16.7	17.0	18.0	18.8	19.6	19.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.7	21.6	21.6	21.6	21.4	20.7	19.2	19.8	20.7	21.2	21.5	21.6
		MCDB	26.4	26.8	26.6	25.8	25.0	25.1	25.1	25.8	26.7	26.0	25.9	25.7
	2%	WB	21.0	20.9	21.0	21.0	20.7	20.0	18.3	19.2	20.1	20.7	20.9	20.9
		MCDB	25.8	26.2	26.1	25.4	24.7	24.6	24.9	25.2	25.8	25.4	25.3	25.4
	5%	WB	20.5	20.4	20.6	20.6	20.3	19.4	17.8	18.5	19.5	20.1	20.5	20.5
		MCDB	25.4	25.7	25.6	24.8	24.1	23.8	24.9	24.4	24.9	24.8	24.8	24.9
	10%	WB	20.0	19.9	20.1	20.1	19.8	18.8	17.2	18.1	19.1	19.6	20.0	20.0
		MCDB	25.1	25.1	25.1	24.1	23.5	23.2	24.8	23.7	24.1	24.1	24.2	24.4
Mean Daily Temperature Range	5% DB	MDBR	10.8	10.8	10.0	8.6	8.6	9.8	10.8	10.8	10.6	10.2	9.4	9.6
		MCDBR	12.8	13.1	12.0	10.0	9.8	11.0	12.1	12.2	12.0	11.7	10.9	11.4
		MCWBR	4.4	4.4	4.5	4.0	3.6	3.6	3.9	3.8	3.7	4.0	4.3	4.2
	5% WB	MCDBR	10.8	10.8	10.6	9.3	8.8	9.5	10.8	10.4	10.7	10.2	9.9	10.0
		MCWBR	4.3	4.2	4.3	4.0	3.7	3.6	3.7	3.3	3.8	4.2	4.4	4.1
Clear-Sky Solar Irradiance	taub		0.459	0.446	0.401	0.354	0.370	0.408	0.457	0.456	0.411	0.385	0.368	0.408
	taud		2.167	2.215	2.384	2.544	2.464	2.310	2.127	2.145	2.297	2.374	2.453	2.329
	Ebn at Noon		877	891	922	945	904	854	816	839	900	936	957	919
	Edh at Noon		157	152	127	105	109	125	151	154	136	128	117	132
All-Sky Solar Radiation	RadAvg		4.86	5.05	4.85	4.64	4.67	4.95	5.24	5.04	5.04	4.82	4.50	4.61
	RadStd		0.38	0.28	0.27	0.29	0.20	0.33	0.32	0.35	0.28	0.28	0.23	0.32

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page